Semiconductor Electronics: Materials, Devices and Simple Circuits

SHORT NOTES

Semiconductors

(i) Semiconductors behave both like conductors and insulators they behave like conductors when temperature is increased. They behave like insulators at low temperature i.e. 0 Kelvin (0 K) Ex. silicon and Germanium.

(ii) In all solids, different energy levels combine to form two bands.

(iii) Valence band (*VB*) contains valence electrons which may be partially or fully filled.

(iv) Conduction band (*CB*) contains free electrons which may be empty or partially filled.

(v) The energy difference between valence band and conduction band is called forbidden energy gap.

(vi) The forbidden energy gap is less for conductors and large for insulators and in between for semiconductors.

(vii) Pure semiconductors are called intrinsic semi conductor Ex. Silicon and Germanium (Si and Ge).

P-N Junctions Diode

(i) A *p*-*n* junction is a single piece of semiconductor one half of which is *p*-type and the other half is *n*-type.

(ii) The region near the junction is called depletion layer.

(iii) There are two types of connection of diode

(*a*) Forward bias (*b*) Reverse bias

(iv) When *p*-type is connected to positive and *n*-type connected to negative terminal then it is forward biased.

(v) In forward bias it offers minimum resistance, depletion region is narrowed. It is similar to 'on' in an electrical switch.

(vi) In reverse bias *p*-type is connected to negative terminal and *n*-type is connected to positive terminal

(vii) In reverse bias it offers maximum resistance, does not conduct and depletion region is widened. It is similar to 'off' in an electrical switch.

Zener Diode

(i) A properly and highly doped *p*-*n* junction diode which operates in reverse bias condition is called 'Zener diode'.

(ii) Silicon is preferred for making Zener diodes.

(iii) Zener diode is operated in reverse bias, which operates at a voltage called 'Zener voltage'.

(iv) Zener diode is used as a 'Voltage regulator'.

Important Formula

(i) Rectifier efficiency $(\eta) = \dfrac{\text{dc output power}}{\text{ac input power}}$

(ii) Half wave rectifier efficiency $(\eta) = \dfrac{0.406 \times R_L}{r_f + R_L}$

r_f = diode forward resistance, R_L = load resistance

(iii) Full wave rectifier efficiency $\eta = \dfrac{0.812 \times R_L}{r_f + R_L}$